



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
Instruction Division

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

INSTRUCTION DIVISION

FIRST SEMESTER 2026-2027

COURSE HANDOUT(PART-II)

Date: 02-08-2026

In addition to Part-I (General Handout for all courses appended to the Timetable) this portion gives further specific details regarding the course.

Course Number: MAC F312

Course Title: Foundations of Data Science

Instructor In Charge: Prof. NILADRI CHATTERJEE ([niladri.chatterjee@pilani.bits-pilani.ac.in](mailto:niladri.Chatterjee@pilani.bits-pilani.ac.in))
Chamber number : 6069H

Course Description: This course introduces the mathematical foundations of data science with emphasis on probability, Gaussian models, statistical inference, nonparametric methods, optimization, dimensionality reduction, data preparation, and scalable data analytics. It develops core tools such as probability bounds, Bayesian inference, PCA and SVD, kernel density estimation, nearest-neighbor methods, and connects them to practical data-analysis problems.

Pre-requisites: The course is self-contained. However, a basic understanding of probability theory, calculus, linear algebra, and elementary programming would be helpful.



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Course Objectives: Students will learn to

- Model data using mathematically grounded ideas from probability, statistics, and linear algebra.
- Bayesian updating, mixture models, nonparametric density and neighborhood methods.
- Deal with high-dimensional data, PCA and SVD, and basic scalable analytics.
- Support data preprocessing, visualization, and interpretation in modern data-science workflows.

Text Book.

[T1] Christopher M. Bishop, “Pattern Recognition and Machine Learning”, Springer, 2006.

[T2] Avrim Blum, John Hopcroft, and Ravindran Kannan, “Foundations of Data Science”, Cambridge University Press, 2020.

Reference Books.

[R1] Jiawei Han, Micheline Kamber and Jian Pei, “Data Mining Concepts and Techniques”, 3rd Edition, Morgan Kaufmann



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Course Plan.

Module	Topics	Lecture	References	Learning Objectives
M I: Foundations and Scope of Data Science	Introduction to Data Science. Revisit Basic Probability Theory. Transformations of Random Variables. Gaussian Distribution, Conditional Gaussian, Marginal Gaussian Distribution, Mixture of Gaussian distributions Generating Functions: MGF, PGF and properties. Distributions derived from Gaussian: Chi Sq, Students t, Introduction to Decision Theory Introduction to Information Theory	L1-L8	[T1]: Ch 1, Ch 2 Class notes	Students will understand the data-science pipeline, the role of mathematical modelling, the role of probability and foundational concepts





Module	Topics	Lecture	References	Learning Objectives
M II:	Bayesian Probabilities. Basics: Prior Probability, Posterior Probability, Likelihood, Conjugate Priors. Bayesian Curve Fitting Advanced: Bayesian Updating, Posterior Mean, Posterior variance, Predictive Distributions,	L9-L14	[T1]: Ch 1, Ch 2	Students will learn Bayesian method of inferencing, importance of priors and posteriors.
M III: Nuances of High Dimensional Space	Law of Large Numbers, Geometry of large numbers, Generating Uniform points in the unit ball. Gaussians in High Dimension. Curse of Dimensionality, Best-Fit Subspaces and Singular Value Decomposition (SVD), Principal Component Analysis, Applications	L15-L20	[T1]: Ch 1; [T2] Ch 3	Laws of large numbers required for data-science models.





Module	Topics	Lecture	References	Learning Objectives
MIV Nonparametric Methods	Nonparametric Statistical Methods: Rank Test, Ratio Test, Kendall-Tau, Wilcoxon, Mann-Whitney, Kruskal-Wallis, Chi-Square Test, Kolmogorov-Smirnov Test, Kernel Density Estimators, Nearest Neighbour Methods	L21-L24	Instructor notes + [T1]: Section 2.5	Students will compare classical nonparametric tests and understand when distribution-free methods are useful.
MV Models for Regression	Linear vs. nonlinear regression, Linear basis function models, MLE and Least Squares, Regularized Least squares,	L25-L28	[T1]: Ch 3	Students will learn mathematics of regression analysis,
MVI Models for Classification	Unsupervised Learning - Clustering, KNN, Logistic Regression, Support Vector Machine, Decision Trees, Random Forests	L29-L34	[R1]Ch 8,9, 10 + Lecture Notes	Students will learn fundamental classification techniques of Machine Learning





Module	Topics	Lecture	References	Learning Objectives
MVII Neural Networks	Perceptron, Multilayer Perceptron, Feedforward Neural Network, Back propagation, Network Training, Regularization in Neural Networks, Types of Neural Networks, Deep Learning Models	L34-L40	[T1]: Ch 5 [T2] Ch 5 + Class notes	Students will learn basics of Neuro computing, parameter optimization, Gradient Descent optimization, Regularization in Neural Networks
Total Lectures		40		

Evaluation Plan.





Details	Duration	Weightage (%)	Date and Time
Quiz-Q1 (Closed Book)	15 mins	5	TBA
Quiz-Q2 (Closed Book)	15 mins	5	TBA
Quiz-Q3 (Closed Book)	15 mins	5	TBA
Quiz Best 2 of 3 (10%)		10	--
Lab Assignment Lab (10%)	Throughout Semester	10	TBA
Self Study Report (20%)	October 28, 2026	20	
Mid-Semester Exam (Closed Book) Midsem (25%)	1.5 Hours	25	10 October FN1
Comprehensive Examination (Closed Book) Compre (35%)	3 Hours	35	15 December FN





Chamber Consultation Hour: TBA. Through Email appointment

Make-Up Policy: No makeup will be given for the quizzes. Make-up for other components will be given only in genuine cases of absence as per institute rules. Prior permission from I/C is a must for makeup of any evaluation component.

NC Criteria: Minimum of 10 % of the total marks of the course.

Attendance Policy: Students are expected to attend all the lectures and tutorials. The attendance is not linked to any of the academic evaluation components.

Notices: All announcements related to the course will be broadcasted through e-mails.

Note: Self-study and discussion components will be used to connect the mathematical theory with data visualization, multidimensional analysis, big-data workflows, and social-media data. Students are expected to read the assigned sections before these discussion sessions.

Prof Niladri Chatterjee

Instructor-In-Charge



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